



Execution Procedure

Team Oriented Project

Group A

Linux Kernel Extension for DM-Verity Authentication, Verification and Setup

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Cooperation with:

**BMW
GROUP**



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Running with the original rootfs

- **Navigate to the /build directory.**

- **Creating the artifacts:**

```
chmod +x build_artifacts.sh
./build_artifacts.sh
```

- **Start QEMU:**

```
Navigate to the /bootloaders directory
chmod +x qemu_main.sh
./qemu_main.sh
```

- **Log in:**

```
Username: root
Password: root
```

- **Verify dm-verity inside the rootfs:**

```
dmesg | grep -i verity
```

Now you should be able to see that the Kernel verifies the signature, parses the metadata and created the device mapper successfully.

Running with the docker

- **Navigate to the /src directory.**

- **Step up the docker:**

```
chmod +x docker_setup.sh
./docker_setup.sh (system reboot expected)
```

- **Start the Docker container:**

```
chmod +x docker_run.sh
./docker_run.sh
```

Choose option 1 to run the original rootfs.

- **Log in:**

Username: root

Password: root

- **Verify dm-verity inside the container rootfs:**

```
dmesg | grep -i verity
```

Now you should be able to see that the Kernel verifies the signature, parses the metadata and created the device mapper successfully.

Running the test wrapper

- **Navigate to the /build directory.**
- **Run the test wrapper:**

```
chmod +x run_verity_tests.sh  
./run_verity_tests.sh
```

Running a specific integrity test

- **Navigate to the /build directory.**
- **Run the integrity tests script:**

```
chmod +x integrity_tests.sh  
./integrity_tests.sh
```

- **Run a specific integrity test:**

```
./integrity_tests.sh <test_name>
```

- **Launch QEMU:**

```
chmod +x launch_qemu.sh  
./launch_qemu.sh
```

Select the corrupted image that you created before.

Depending on the test, the kernel may panic, reject boot, or load the rootfs normally.